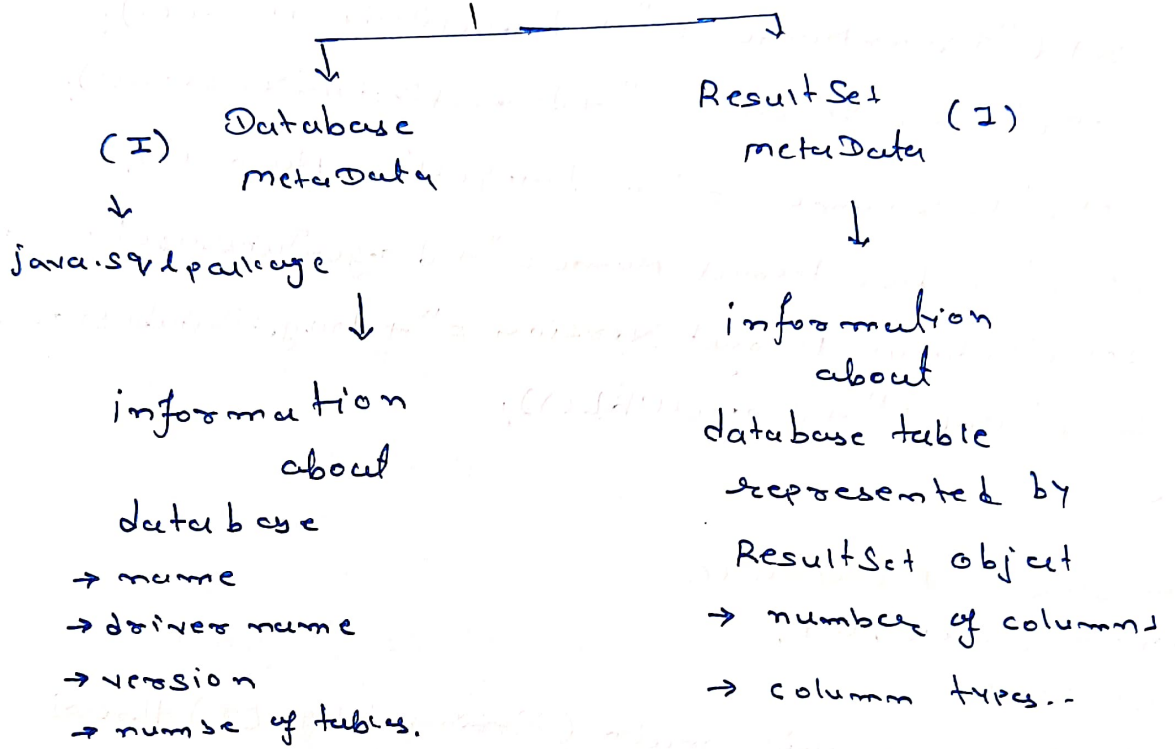


Meta Data

↓
data about data

[Extra Information about our original data]

Types of Meta Data



* Database MetaData.

```
import java.sql.*;
```

```
class DBMD
```

```
{ public static void main (String args[]) throws Exception
```

```
{ // Connection code ....
```

```
DatabaseMetaData dm = con.getMetaData();
```

```
SOP ("Driver Name = " + dm.getDriverName());
```

```
SOP ("Driver Version = " + dm.getDriverVersion());
```

```
SOP ("User name = " + dm.getUserName());
```

```
SOP ("Database Product Name = " + dm.getDatabaseProductName());
```

```
SOP ("Database Product Version = " + dm.getDatabaseProductVersion());
```

```
SOP ("URL : " + dm.getURL());
```

```
}
```

```
}
```

* ResultSet metaData.

```
import java.sql.*;
```

```
class RSMMD
```

```
{ public static void main (String args[]) throws Exception
```

```
{ // Connection Code ...
```

```
→ ResultSet rs = st.executeQuery ("select * from student");
```

```
ResultSetMetaData rsmd = rs.getMetaData();
```

```
SOP ("Table Name = " + rsmd.getTableName());
```

```
int count = rsmd.getColumnCount();
```

```
for (int i = 0; i < count; i++)
```

```
{ SOP ("Column Number : " + (i+1));
```

```
SOP ("Column Name : " + rsmd.getColumnLabel(i+1));
```

```
SOP ("Column Type : " + rsmd.getColumnTypeName(i+1));
```

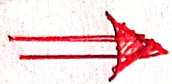
```
SOP ("Column Size : " + rsmd.getColumnSize(i+1));
```

```
SOP ("-----");
```

```
}
```

```
}
```

```
}
```



Working with Large Objects



Large files (images, videos, audio files, resume etc)



database 

Large objects



Binary Large object



Character Large object

CBLOB)

Collection of binary data

Stored as Single Entity



images, video files, audio files

(CLOB)



collection of character data

Stored as Single Entity



large text files.

*> Insertion Blob type into database.

```
import java.sql.*;  
import java.io.*;
```

```
class App
```

```
{ public static void main(String args[]) throws Exception
```

```
{
```

```
    // Connection code....
```

```
    String sql = "create table Actor (name varchar(30), image Blob);"
```

```
    Statement st = con.createStatement();
```

```
    st.execute(sql);
```

```
    sql2 = "insert into Actor (1, 1)";
```

```
    PreparedStatement pst = con.prepareStatement(sql2);
```

```
    File f = new File("D:\\Kartik.jpg");
```

```
    FileInputStream fis = new FileInputStream(f);
```

```
    pst.setString(1, "Kartik");
```

```
    pst.setBlob(2, fis);
```

```
    int r = pst.executeUpdate();
```

```
    if (r > 0)
```

```
        System.out.println("Record updated");
```

```
    else
```

```
        System.out.println("Failed");
```

```
}
```

```
}
```

*> Insertion Clob type into db.

```
import java.sql.*;  
import java.io.*;  
class App  
{  
    public static void main (String args[]) throws Exception  
    {  
        // Connection Code...
```

```
String sql = "create table dataR (name varchar(30), resume longtext)";
```

```
Statement st = con.createStatement();
```

```
st.executeUpdate(sql);
```

```
File f = new File("D:\\PAT-res.txt");
```

```
FileReader fr = new FileReader(f);
```

```
String sql2 = "insert into dataR values (?, ?)";
```

```
PreparedStatement pst = con.prepareStatement(sql2);
```

```
pst.setString(1, "PAT");
```

```
pst.setClob(2, fr);
```

```
int r = pst.executeUpdate();
```

```
if (r > 0)
```

```
    System.out.println("Row Updated");
```

```
else
```

```
    System.out.println("Failed");
```

```
}
```

```
}
```


Retrive clob data from Db.

```
import java.sql.*;
import java.io.*;

class AP
{
    // Connection code....

    File f = new File("G:\\xyz.txt");
    FileWriter fw = new FileWriter(f);

    String sql = "select resume from dataR where name = 'Pri'";
    PreparedStatement pst = con.prepareStatement(sql);

    ResultSet r = pst.executeQuery();

    while (r.next())
    {
        Reader c = r.getCharacterStream(1);
        int i = c.read();
        while (i != -1)
        {
            fw.write((char) i);
            i = c.read();
        }
        fw.close();
    }
}
```

⇒ Retrieve Blob data from db.

```
import java.sql.*;
```

```
import java.io.*;
```

```
class App
```

```
{ public static void main (String args[]) throws Exception
```

```
{ // Connection Code....
```

```
File File f = new File ("D:\\abc.jpg");
```

```
FileOutputStream fos = new FileOutputStream (f);
```

```
Sql = "Select image from Actor where name = 'Kastur';";
```

```
PreparedStatement pst = con.prepareStatement (sql);
```

```
ResultSet r = pst.executeQuery();
```

```
while (r.next())
```

```
{
```

```
InputStream bi = r.getBinaryStream (1);
```

```
InputStream bi = r.getBinaryStream (1);
```

```
int i = bi.read();
```

```
while (i != -1)
```

```
{ fos.write(i);
```

```
i = bi.read();
```

```
}
```

```
fos.flush();
```

```
}
```

```
}
```

```
}
```